Phase 3.1 — Runtime Architecture Design

Meeting 1 — Project Definition and Design Philosophy

Project: Digital Harmonium Workstation
Document Version: 1.0
Status: Finalized
Phase: 3.1 Runtime Architecture Design

---

1. Purpose of this Meeting

The purpose of the first Runtime Architecture meeting is to define the identity, purpose, philosophy, and scope of the Digital Harmonium Workstation before any runtime software architecture is designed.

No firmware implementation decisions are made during this meeting.

Instead, this meeting establishes the design principles that every future hardware and software decision must follow.

This document serves as the foundation upon which the entire runtime architecture will be constructed.

---

2. Why This Project Exists

The Digital Harmonium Workstation began from a simple observation:

Many electronic keyboards eventually become unusable due to damaged electronics while their keybeds, enclosures, and mechanical components remain perfectly functional.

Rather than allowing such instruments to become electronic waste, this project demonstrates that they can be transformed into modern embedded musical workstations using low-cost hardware and professionally designed firmware.

The project therefore has multiple motivations.

2.1 Educational Motivation

To demonstrate the complete design of a professional embedded musical instrument including:

- Embedded systems engineering
- Digital audio processing
- Binary file format design
- Sample-based synthesis
- Real-time firmware architecture
- Low-latency audio systems
- Resource-constrained embedded software

The project is intended to be a complete educational reference for understanding how commercial digital musical instruments operate internally.

2.2 Musical Motivation

The workstation should not merely demonstrate technology.

It should become a genuinely useful instrument for musicians.

The primary objective is to support Indian Classical Music practice through:

- Harmonium
- Tanpura
- Tabla
- Recording
- Low-latency performance
- Multiple synthesized voices

The instrument should feel natural to play rather than functioning only as a demonstration prototype.

2.3 Sustainability Motivation

The project promotes hardware reuse by extending the life of discarded or non-functional electronic keyboards.

Instead of replacing an entire instrument, existing mechanical hardware can be repurposed into a modern embedded musical workstation.

This reduces electronic waste while dramatically lowering overall project cost.

2.4 Accessibility Motivation

Commercial Indian Classical practice equipment is often expensive.

A musician may require separate devices such as:

- Harmonium
- Electronic Tanpura
- Electronic Tabla
- Additional accompaniment devices

Purchasing these separately can become prohibitively expensive for many students.

The Digital Harmonium Workstation aims to integrate these capabilities into a single affordable embedded instrument while maintaining practical musical usefulness.

---

3. Product Identity

The Digital Harmonium Workstation is defined as:

> **A standalone embedded musical practice instrument designed primarily for Indian Classical Music that provides synthesized melodic voices, accompaniment, recording, and real-time performance on low-cost hardware.**

This definition intentionally avoids describing implementation details such as processor type, programming language, or electronics.

It defines what the product is rather than how it is implemented.

---

4. Core Mission

The mission of the project is:

> **To create a professional-quality embedded musical instrument that is both educational for engineers and genuinely useful for musicians.**

Neither objective is considered secondary.

The workstation must satisfy both simultaneously.

---

5. Target Users

The workstation is designed for three primary categories of users.

5.1 Embedded Systems Student

This user studies the project to understand:

- Embedded software architecture
- Audio firmware
- DSP concepts
- Binary serialization
- Real-time scheduling
- Resource management

For this user, the firmware itself serves as an educational reference.

5.2 Musician

The musician is interested only in using the instrument.

Their priorities include:

- Low latency
- Reliable performance
- Authentic musical experience
- Practical accompaniment
- Recording capability

The musician should not need to understand the internal implementation.

5.3 Future Developer

Future developers may extend or maintain the project.

Therefore the software architecture must remain:

- Modular
- Readable
- Maintainable
- Well documented
- Logically organized

The architecture should remain understandable years after its original development.

---

6. What This Project Is Not

Clearly defining project boundaries is essential for preventing uncontrolled feature growth.

The Digital Harmonium Workstation is **not** intended to become:

- A Digital Audio Workstation (DAW)
- A VST Instrument
- A Commercial Yamaha Workstation Replacement
- A General-Purpose Sampler
- A Studio Production Platform
- A Consumer Entertainment Device
- A Feature-Maximized Keyboard

Features unrelated to the core mission will not be included merely because they are technically possible.

---

7. Core Identity Philosophy

The most important design philosophy established during this meeting is:

> **The workstation is first and foremost a musical instrument.**

Every future feature shall be evaluated according to a single question:

> **Would a musician genuinely use this feature during musical practice or performance?**

If the answer is yes, the feature supports the project.

If the answer is no, the feature should be reconsidered regardless of its technical appeal.

This philosophy ensures that engineering decisions always remain aligned with musical usefulness.

---

8. Design Principles

The following principles govern all future design decisions.

Principle 1

Musical Usability Over Technical Perfection

The primary objective is to create an instrument that feels convincing and enjoyable to play.

Perfect physical simulation is not required if a simpler solution produces an equally satisfying musical experience.

Principle 2

Real-Time Performance Is Sacred

Audio stability is considered the highest technical priority.

No feature shall compromise:

- Audio continuity
- Latency
- Timing
- Reliability

Real-time behavior always takes precedence over additional functionality.

Principle 3

Simplicity Over Cleverness

The architecture should favor:

- Readability
- Maintainability
- Predictability

Complex solutions should only be adopted when they provide clear practical advantages.

Principle 4

Every Feature Must Justify Its Cost

ESP32 resources are finite.

Every proposed feature must justify:

- CPU usage
- RAM usage
- Flash storage
- Development complexity

Features without sufficient practical value should not be included.

Principle 5

The VHL Format Is the Single Source of Truth

All runtime behavior should derive directly from VHL data.

Hidden configuration files, undocumented runtime behavior, or duplicated parameter definitions should be avoided.

The VHL file represents the complete instrument specification.

Principle 6

The Architecture Must Be Educational

The firmware should itself serve as an educational reference.

A reader should be able to understand:

- Runtime architecture
- Voice allocation
- DSP pipeline
- Scheduling
- Embedded software organization

without requiring extensive external explanation.

Principle 7

Every Feature Must Support the Core Identity

The workstation is a musical practice instrument.

Features shall therefore be included only if they directly support musical practice or performance.

Examples include:

- Harmonium voices
- Tanpura
- Tabla
- Recording
- Transpose
- Tempo control
- Low-latency playback

Features unrelated to the instrument's identity should be excluded.

---

9. Architecture Freeze Policy

The completion of this meeting establishes the permanent philosophical foundation of the project.

Subsequent meetings may define:

- Hardware
- Runtime architecture
- Memory organization
- Tasks
- Scheduling
- Audio processing

However, they must remain consistent with the principles established in this document.

Future design decisions should reference this document whenever architectural questions arise.

---

10. Meeting Outcome

The following items have been finalized:

- Project motivation
- Product identity
- Core mission
- Target users
- Project scope
- Design philosophy
- Design principles
- Architecture freeze policy

These decisions collectively define the long-term direction of the Digital Harmonium Workstation and serve as the conceptual foundation for all subsequent runtime architecture design.

Next Meeting

Phase 3.1 — Meeting 2

Feature Freeze and Hardware Definition

This meeting will define:

- Complete feature set
- Hardware components
- Physical controls
- User interface elements
- External connections
- Display requirements
- Input devices
- Final workstation configuration

Only after these decisions are finalized will runtime architecture design begin.